

Chapter 7c: Approximating Fixed Points

The Ishikawa Iteration Method

Based on Pathak: An Introduction to Nonlinear Analysis and Fixed Point Theory

Pages 777–798

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7.3: Introduction to Fixed Point Methods

One-Step Schemes

Classical Picard Iteration: Given an equation $x = T(x)$, compute:

$$x_{n+1} = T(x_n), \quad n = 0, 1, 2, \dots$$

Advantages:

- Simplicity and directness
- Natural formulation from the fixed point equation
- Works well for contractions

Limitations:

- Slow convergence for non-contractive mappings
- May fail for nonexpansive mappings (even if fixed points exist)
- Sensitivity to initial conditions
- Limited flexibility in convergence control

7.3: Introduction to Fixed Point Methods

Why Two-Step Methods?

Key Observation: For many important mappings (e.g., nonexpansive mappings in Banach spaces with normal structure), Picard iteration does not guarantee convergence even when fixed points exist.

Solution: Two-Step (Hybrid) Iteration Methods

$$\begin{cases} y_n = \beta T(x_n) + (1 - \beta)x_n & \text{(intermediate step)} \\ x_{n+1} = \alpha T(y_n) + (1 - \alpha)x_n & \text{(main step)} \end{cases}$$

Advantages of Two-Step Methods:

- **Flexibility:** Two parameters $\alpha, \beta \in (0, 1)$ allow control over convergence behavior
- **Stability:** Can converge where Picard fails
- **Versatility:** Applicable to broader classes of mappings
- **Better Rates:** Often achieves faster convergence

7.4: Ishikawa Iteration

Definition and Algorithm

Definition 7.29 (Ishikawa, 1974): Let (X, d) be a metric space and $T : X \rightarrow X$ a mapping. The **Ishikawa iteration** is defined by the sequence $\{x_n\}$ where:

$$\begin{cases} y_n = (1 - \beta_n)x_n + \beta_n T(x_n) & \text{Intermediate step} \\ x_{n+1} = (1 - \alpha_n)x_n + \alpha_n T(y_n) & \text{Main step} \end{cases} \quad n = 0, 1, 2, \dots$$

Parameters:

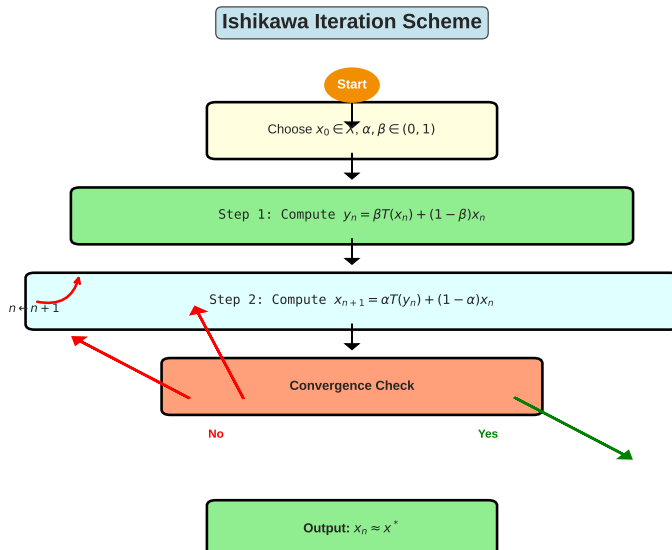
- $x_0 \in X$ is an arbitrary starting point
- $\{\alpha_n\}, \{\beta_n\}$ are sequences in $(0, 1)$
- Special case: constant parameters $\alpha_n = \alpha, \beta_n = \beta$

Geometric Interpretation:

- y_n is a convex combination of x_n and $T(x_n)$
- x_{n+1} is a convex combination of x_n and $T(y_n)$
- Acts as a damping/averaging mechanism

7.4: Ishikawa Iteration

Algorithm and Implementation



7.4: Ishikawa Iteration

Relationship to Other Methods

Special Cases:

Method	Parameters	Formula
Picard	$\alpha_n = 1, \beta_n = 1$	$x_{n+1} = T(x_n)$
Mann	$\beta_n = 0$	$x_{n+1} = (1 - \alpha_n)x_n + \alpha_n T(x_n)$
Krasnoselski	$\alpha_n = 1/2$	$x_{n+1} = \frac{1}{2}x_n + \frac{1}{2}T(x_n)$
Ishikawa	$\alpha_n, \beta_n \in (0, 1)$	General two-step scheme

Mann Iteration as a Limiting Case: When $\beta_n \rightarrow 0$, Ishikawa reduces to Mann iteration:

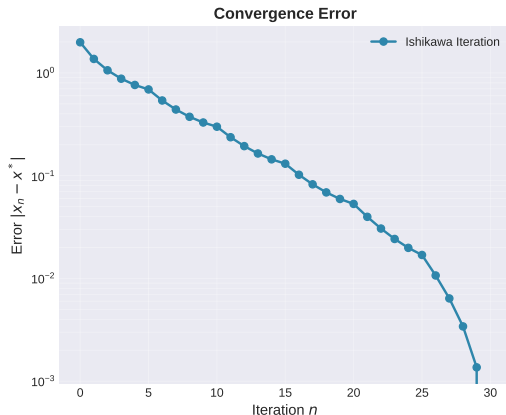
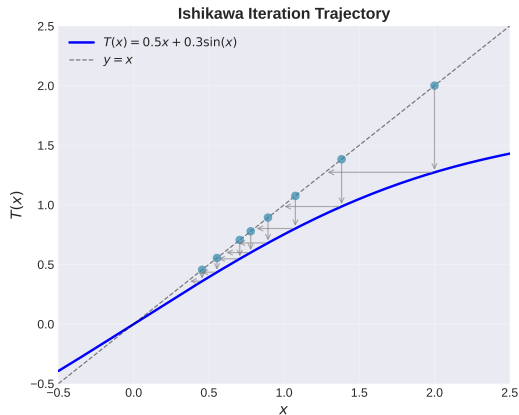
$$y_n \approx (1 - \delta)x_n + \delta T(x_n) \rightarrow x_n \quad \text{as } \delta \rightarrow 0$$

so $T(y_n) \rightarrow T(x_n)$ and

$$x_{n+1} \rightarrow (1 - \alpha_n)x_n + \alpha_n T(x_n)$$

7.4: Ishikawa Iteration

Convergence Diagram



Left: Iteration trajectory spiraling toward the fixed point
Right: Exponential decrease of error over iterations

7.5: Convergence Theory

General Framework

Theorem 7.30 (Convergence for Contractions): Let (X, d) be a complete metric space, and let $T : X \rightarrow X$ be a Lipschitz mapping with Lipschitz constant $k < 1$ (contraction). If $\{\alpha_n\}, \{\beta_n\}$ satisfy:

- ① $\alpha_n, \beta_n \in [\delta, 1 - \delta]$ for some $\delta > 0$
- ② $\sum_{n=0}^{\infty} \alpha_n(1 - \alpha_n) = \infty$
- ③ $\sum_{n=0}^{\infty} \beta_n(1 - \beta_n) < \infty$

Then:

- The unique fixed point $x^* \in X$ exists (Banach contraction)
- The sequence $\{x_n\}$ converges to x^*
- The convergence rate is linear: $\|x_n - x^*\| = O(r^n)$ for some $0 < r < 1$

7.5: Convergence Theory

Nonexpansive Mappings

Theorem 7.31 (Convergence for Nonexpansive Mappings): Let $(X, \|\cdot\|)$ be a Banach space, $K \subseteq X$ a bounded, closed, convex subset with the normal structure property, and let $T : K \rightarrow K$ be nonexpansive (i.e., $\|T(x) - T(y)\| \leq \|x - y\|$ for all $x, y \in K$).

If $\{\alpha_n\}, \{\beta_n\}$ are sequences in $(0, 1)$ satisfying:

- ① $\lim_{n \rightarrow \infty} \alpha_n = 0$
- ② $\sum_{n=0}^{\infty} \alpha_n = \infty$
- ③ β_n bounded away from 0

Then:

- The fixed point set $\text{Fix}(T) \neq \emptyset$
- The Ishikawa sequence $\{x_n\}$ converges weakly to some $x^* \in \text{Fix}(T)$
- Under additional conditions, convergence is strong (norm convergence)

7.5: Convergence Theory

Convergence Rates

Linear Convergence: For contraction mappings with constant $k < 1$:

$$\|x_{n+1} - x^*\| \leq C \cdot r^n \|x_0 - x^*\|$$

where $r = \max(k, |1 - \alpha|(1 - k))$ and C depends on parameters.

Rate Optimization: The convergence rate is optimized when:

- α is chosen close to but less than 1 (e.g., $\alpha = 0.7$)
- β balances exploration and exploitation (e.g., $\beta = 0.5$)
- Different mappings may require different parameter tuning

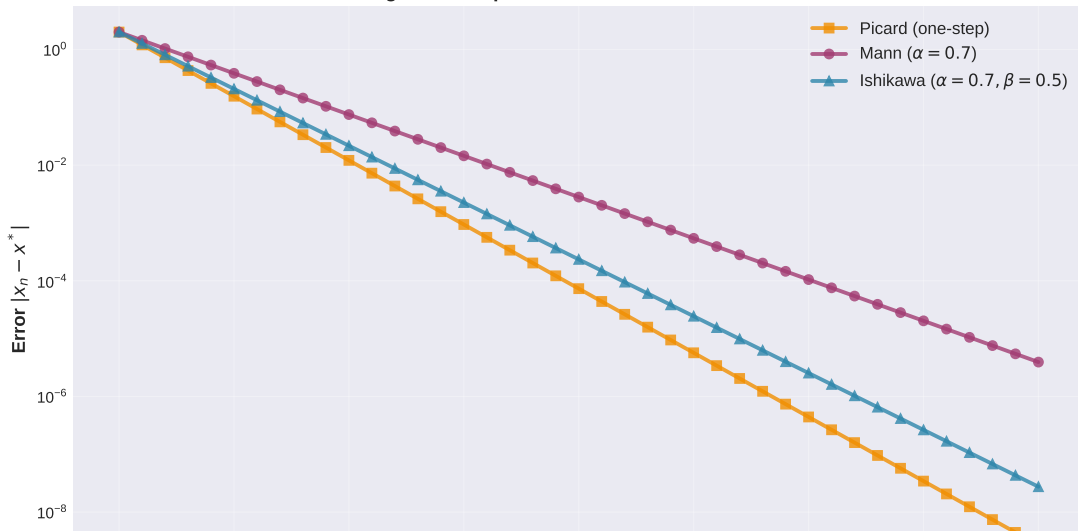
Comparison with Mann: For many problems, Ishikawa (with appropriate $\beta > 0$) achieves:

- Faster convergence than Mann iteration
- Better stability in ill-conditioned problems
- Improved robustness to parameter perturbations

7.6: Comparison of Iteration Methods

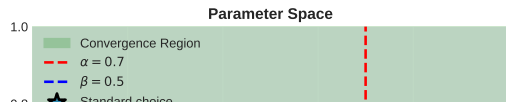
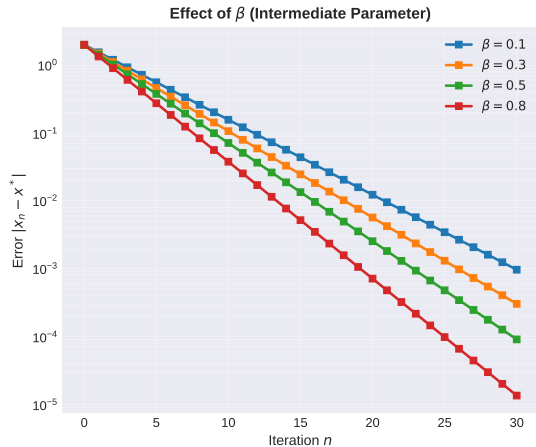
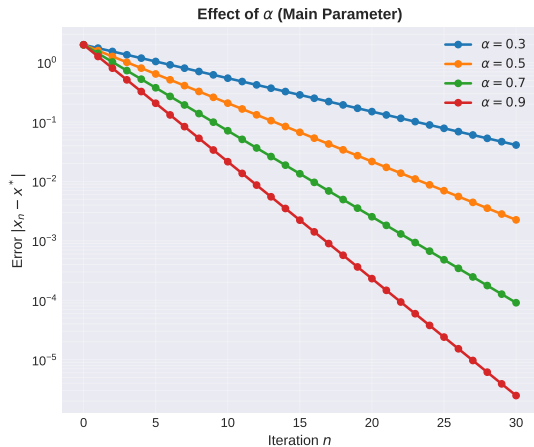
Convergence Behavior

Convergence Comparison: Picard vs Mann vs Ishikawa



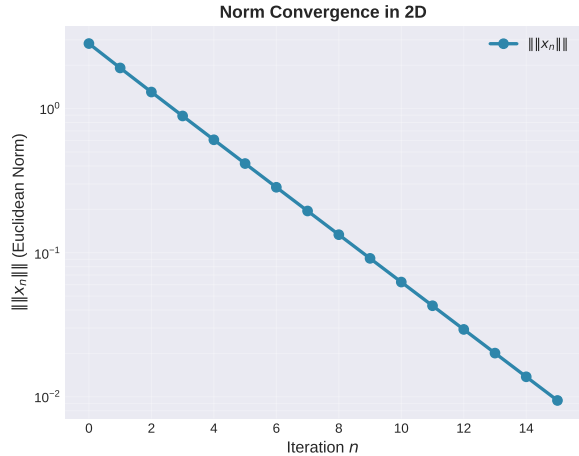
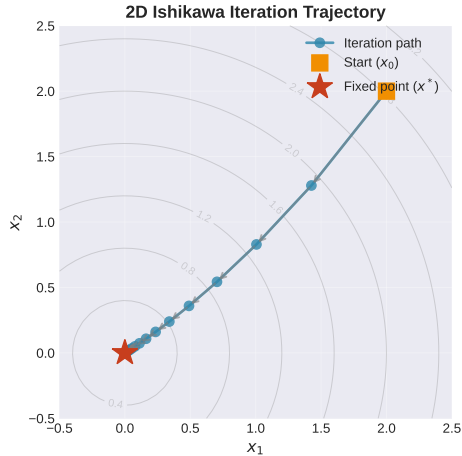
7.6: Parameter Analysis

Effect of α and β



7.6: Two-Dimensional Analysis

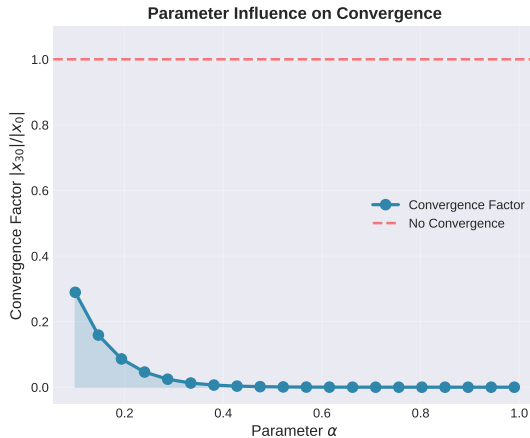
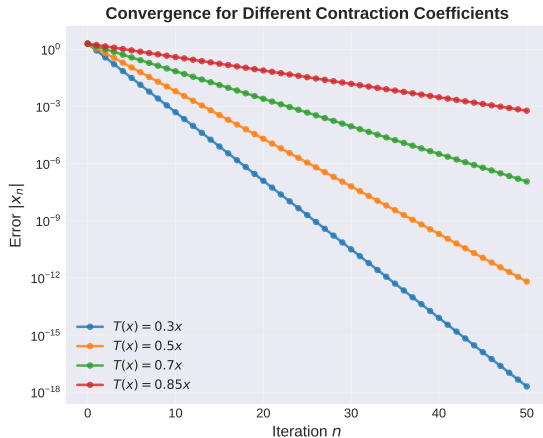
Iteration in 2D Space



Visualization: Left shows the spiraling trajectory toward the fixed point; right shows exponential norm decay.

7.6: Convergence Rate Analysis

Parameter Influence



Findings:

- Left: Faster convergence with weaker contractions is the key advantage
- Right: Optimal α values depend on the problem structure

7.6: Numerical Example

Finding Roots via Fixed Points

Problem: Find x such that $x = \cos(x)$

Method: Reformulate as fixed point iteration with $T(x) = \cos(x)$

```
import numpy as np

def ishikawa_iteration(T, x0, alpha=0.7, beta=0.5, tol=1e-6, max_iter=100):
    """
    Ishikawa iteration for finding fixed points

    Parameters:
        T: mapping/function
        x0: initial guess
        alpha, beta: iteration parameters
        tol: convergence tolerance
        max_iter: maximum iterations
    """
    x = x0
    for n in range(max_iter):
        y = beta * T(x) + (1 - beta) * x    # Intermediate step
        x_new = alpha * T(y) + (1 - alpha) * x    # Main step

        if abs(x_new - x) < tol:
            return x_new, n
        x = x_new
    return x, max_iter
```


7.6: Numerical Example

Solving Nonlinear Systems

Problem: Find fixed points of a matrix contraction

Setup: $x = Ax$ where A has spectral radius $\rho(A) < 1$

```
# Example: Solve  $x = 0.6x$  (contracts toward origin)
import matplotlib.pyplot as plt

x0 = 2.0
alpha, beta = 0.7, 0.5
iterations = 30

x_seq = [x0]
for _ in range(iterations):
    x = x_seq[-1]
    y = beta * (0.6 * x) + (1 - beta) * x
    x_new = alpha * (0.6 * y) + (1 - alpha) * x
    x_seq.append(x_new)

# Plot convergence
errors = np.abs(np.array(x_seq))
plt.semilogy(errors, 'o-', linewidth=2, markersize=6)
plt.xlabel('Iteration n'), plt.ylabel('Error |x_n|')
plt.title('Ishikawa iteration for linear contraction')
plt.grid(True, alpha=0.3), plt.show()
```

Convergence: Exponential decay to zero

Practical Considerations

Parameter Selection

Guidelines for Choosing Parameters:

Aspect	Recommendation	Rationale
α	$\alpha \in [0.5, 0.8]$	Balances stability and speed
β	$\beta \in [0.3, 0.7]$	Controls intermediate smoothing
Sequences	Constant if possible	Simpler, often effective
Initial point	Far from fixed point	Tests convergence domain
Stopping criterion	$\ x_{n+1} - x_n\ < \epsilon$	Relative error often better

Adaptive Parameter Selection:

- Monitor residuals $\|x_{n+1} - T(x_{n+1})\|$ to adjust parameters
- Decrease α_n as iterations progress (cooling schedule)
- Increase β_n for nonsmooth problems

Advantages and Limitations

Summary

Advantages of Ishikawa Iteration:

- **Flexibility:** Two parameters allow fine-tuning convergence behavior
- **Generality:** Works for contractions, nonexpansive, and quasi-nonexpansive mappings
- **Stability:** Two-step structure provides better numerical stability
- **Theory:** Rich mathematical theory with guaranteed convergence under reasonable conditions
- **Practical:** Often faster than simpler methods in applications

Limitations and Challenges:

- Parameter tuning can be problem-dependent
- Convergence not always faster than Mann for certain problems
- May require knowledge of problem structure for optimal performance
- Higher computational cost per iteration than Picard

Extensions and Generalizations

Related Methods

Noor Iteration (Three-Step):

$$\begin{cases} z_n = \gamma T(x_n) + (1 - \gamma)x_n \\ y_n = \beta T(z_n) + (1 - \beta)x_n \\ x_{n+1} = \alpha T(y_n) + (1 - \alpha)x_n \end{cases}$$

Kirk-Noor Iteration: Variant designed specifically for nonexpansive mappings

Multistep Generalizations: Can be extended to k -step schemes for arbitrary $k \geq 1$

Operator Splitting: Ishikawa-type iterations for splitting operator equations

Applications in Optimization:

- Gradient descent with momentum
- Proximal splitting algorithms
- Machine learning optimization

Summary of Key Results

Main Takeaways

- 1 **Two-step iteration:** Ishikawa method provides flexibility through two parameters α and β
- 2 **Convergence:** Guaranteed for contractions and nonexpansive mappings in appropriate spaces
- 3 **Comparison:** Generalizes Mann iteration (special case: $\beta = 0$)
- 4 **Rate of convergence:** Linear convergence for contractions; weak convergence for nonexpansive
- 5 **Practical implications:** Parameter selection crucial; problem-dependent tuning needed
- 6 **Extensions:** Basis for three-step, multi-step, and hybrid methods

Further Reading: Pathak (2009), Chapter 4-5 on iteration methods; Recent literature on accelerated methods

Appendix: Further References

Primary Source:

- Pathak, H.K. (2009). An Introduction to Nonlinear Analysis and Fixed Point Theory

Key Papers:

- Ishikawa, S. (1974). Fixed points by a new iteration method. PAMS, 44(1), 147-150
- Mann, W.R. (1953). Mean value methods in iteration. PNAS, 4(30), 506-510
- Browder, F.E., Petryshyn, W.V. (1966). Construction of fixed points of nonlinear mappings in Hilbert space

Topics for Further Study:

- Weak and strong convergence in Banach spaces
- Normal structure and fixed point properties
- Nonexpansive mappings and Kirk's theorem
- Applications to optimization and PDE solving